

Tuscany Pyroconvection-Type Forecast – ICON-2I 2.2 km – VALID 2026-07-14

3-hourly, 24 h. Run 2026-07-14 00Z. Method after Castellnou et al. (2022), JGR-Atmos.

How to read this product

Two panels are produced. The **fuel-gated** map is the expected, operationally comparable product (the equivalent of the Catalan “tipus de piroconveccio” map): a pyroCu/pyroCb class is assigned **only where a fire could actually reach ≥ 10 MW/m** of fireline intensity on the real Tuscany fuels. The **potential** map is an unconditional **upper bound** – it assumes a pyroCu-capable fire in *every* cell. Use the gated panel for situational awareness; use the potential panel only to see the atmospheric ceiling.

Method and its limits (read this before using the classes)

The column is classified from a **real vertical profile**, not a standard atmosphere: per-cell heights come from the model geopotential (FI), the mixing depth from the **bulk Richardson number** (first Rib ≥ 0.33 above 200 m AGL, referenced to the 2 m / 10 m state), the LCL from the exact **Bolton (1980)** formula, and the humidity at the ABL top from the model’s RH. The cap gamma-theta is taken over ABL+200 m to ABL+1200 m.

The **mixed-layer stability** diagnostic is the weakest link in this product, and is a *proxy*, not a measurement. Validated against IGRA radiosondes (JJA 12Z, period of record, n=2835), a 5-pressure-level column contains only 0-2 model levels inside the mixed layer – one or none in 92% of coastal columns – and the lowest sits in the superadiabatic surface layer. The mixed-layer dtheta/dz is therefore **not measurable from this archive**: fitting it across the in-mixed-layer levels has no skill (Youden J ~ 0.00), and measuring theta up to the Rib ABL top folds in the entrainment jump (Delta-theta), which Castellnou et al. (2022, sec.2.1.1) treat as a variable *separate* from the gradient the ladder conditions on.

What is used instead is the **parcel mixing depth** as a proxy: at the $1.1\text{e-}3$ K/m threshold the criterion reduces to “well-mixed layer ≥ 455 m deep”. It is the only candidate with skill (J = 0.29 inland / 0.55 coastal, r = +0.50 against the radiosonde truth). It **over-flags**: inland specificity is 0.29, so of the columns that are truly stable it still calls $\sim 71\%$ pyroCu-capable (sensitivity 0.99 – it rarely misses a capable column).

Consequence: read the classes as a screening flag, not as calibrated counts. The class *totals* on these maps are not quantitatively trustworthy. Resolving this requires more vertical levels (ERA5’s 37 pressure levels, or ICON native model levels), which the open-data archive does not publish. Set PYROCONV_ML_METHOD=surface_to_abl or mid_layer for the superseded measurements.

Ladder actually used for this run: noshear (5 usable pressure levels). The full method has a fifth diagnostic – the distance from the ABL/LCL to the height of maximum wind shear – but locating a shear *maximum* needs a resolved profile, and the ICON-2I open-data archive publishes only 5-6 pressure levels. The classifier therefore **drops the shear test rather than invent one** from an interpolant; the shear clause is a *necessary* condition for the top class, so omitting it makes class 4 somewhat **easier** to reach here than in the full method. Treat class 4 as an alert to inspect the column, not as a calibrated probability.

The classes express atmospheric predisposition **given a fire of sufficient power**; in the gated panel that power is computed, not assumed. They are paper-informed thresholds, not locally validated ones.

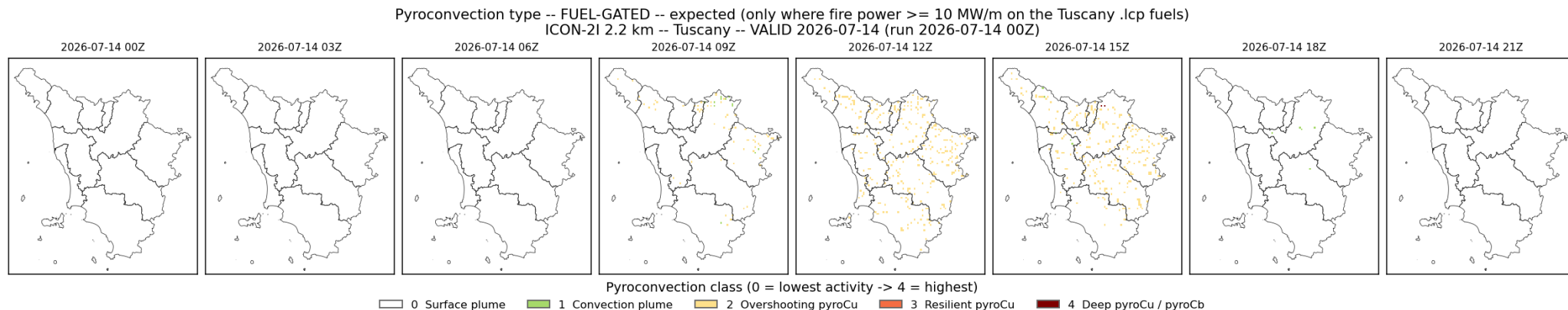


Figure 1: gated

Expected pyroconvection type – FUEL-GATED

Atmospheric potential – UPPER BOUND (assumes a pyroCu-capable fire in every cell)

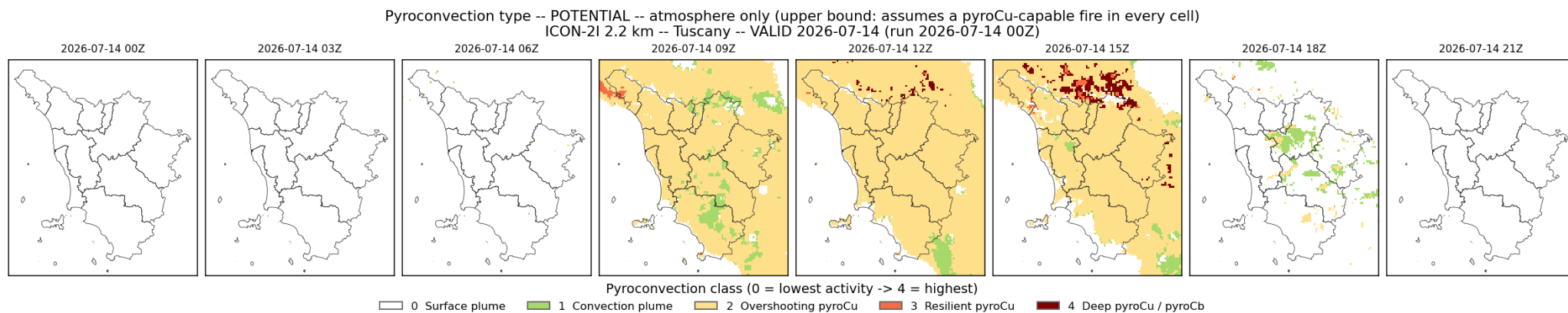


Figure 2: potential

Class scale (low -> high pyroconvective activity)

Level	Colour	Class	Meaning
0	white	Surface plume	Buoyant smoke plume; no significant cloud development.

Level	Colour	Class	Meaning
1	green	Convection plume	Plume penetrates a stable mixed layer; condensation possible, no pyroCu.
2	yellow	Overshooting pyroCu	Brief pyrocumulus; cloud base above the mixing height (LCL/ABL > 1).
3	orange	Resilient pyroCu	Persistent pyrocumulus in an unstable column (LCL/ABL < 1).
4	dark red	Deep pyroCu / pyroCb	Deep pyroconvection / pyrocumulonimbus; weak upper cap lets the plume deepen.

Classification thresholds (ladder in use: **noshear**)

Diagnostic	Threshold	Effect
ML stability <i>proxy</i> : parcel mixing depth (see Method)	depth < 455 m (“stable”)	Convection plume only – no pyroCu
ML stability proxy	depth >= 455 m (= dtheta/dz <= 1.1e-3 K/m)	Column is pyroCu-capable (weak filter: spec. 0.29 inland)
LCL / ABL ratio	1.0 – 1.60	Overshooting pyroCu (brief)
LCL / ABL ratio	< 1.0	Resilient pyroCu (persistent)
LCL / ABL ratio	<= 1.10 (+ conditions below)	Admissible for deep pyroCu / pyroCb
Cap gamma-theta (ABL+200 m -> ABL+1200 m)	<= 4.2e-3 K/m (weak cap)	Permits deepening to pyroCb
Cap gamma-theta	>= 4.8e-3 K/m (strong cap)	Inhibits deepening (resilient at most)
RH at the ABL top (mean, ABL +/- 150 m)	>= 80%	Required for classes 3 and 4
Shear-maximum distance / ABL	<= 0.30	Required for class 4 in the 5-diagnostic ladder only (not resolvable here – see Method)
Fireline intensity (fuel gate, gated panel)	>= 10 MW/m	Minimum fire power for any pyroCu (Tedim et al. 2018)
ABL depth	< 600 m	Held at surface plume (mixing too shallow)
Usable pressure levels	< 4	Cell not classified

Reference cases (Castellnou et al. 2022, Table 1)

The paper’s labelled events, which anchor the published 3-diagnostic ladder (`pyflam.pyroconvection_type(ladder="castellnou")`), still the library default and regression-tested against these rows). The profile ladders used for this map are stricter at the top: they additionally require a moist ABL top and an LCL close to it.

Case	Observed type	LCL/ABL	ML dtheta/dz	gamma-theta (700-500)
T21	Convection plume	–	stable	–
SCQ32	Overshooting pyroCu	> 1	neutral/unstable	–
M11	Resilient pyroCu	< 1	unstable	4.2e-3 (resilient cap)
SCQ41	pyroCu, not pyroCb	< 1	unstable	5.1e-3 (strong cap)
SCQ51	Deep pyroCu / pyroCb	< 1	unstable	3.9e-3 (weak cap)

Forcing: ICON-2I 2.2 km full-Italy GRIB (MISTRAL / AgenziaItaliaMeteo, CC-BY), Tuscany subset: geopotential, temperature, RH and wind on 5 pressure levels plus the 2 m / 10 m state, surface pressure and orography. Classifier: pyflam.pyroconvection_type (bulk-Richardson ABL, Bolton LCL, mixed-layer $d\theta/dz$, cap γ - θ , ABL-top RH; no surface CAPE). Gate: Rothermel + Cruz-2005 crown on the .lcp fuels with forecast moisture/wind. Per-hour diagnostic rasters (ABL, LCL, LCL/ABL, ML $d\theta/dz$, cap, RH-top) accompany the classes. Province borders: ISTAT-derived (openpolis geojson-italy). Generated by tests/pyroconv_daily.py.